SESSION-5
MODULE-1(TASK)

1. **Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.**

ANS:

- Initialize a local repository: git init
- Stage all files: git add .
- Commit with a message: git commit -m "Your commit message"
- Push to a remote repository: git push origin main *(or your specific branch name)*
- Pull latest changes: git pull

2. **What is the difference between git fetch and git pull? Answer in 2 lines.**

ANS:

git fetch securely downloads the latest commits, branches, and tags from a remote repository into your local copy without altering your current working files. git pull goes a step further by immediately merging those downloaded changes directly into your active local branch, combining a git fetch and a git merge into a single action.

3. **Create a new GitHub repository, push any small text file from your local system, and share the repo link.**

ANS:

1. Create a new directory and move into it

mkdir my-new-repo && cd my-new-repo

2. Initialize Git and create a small text file

git init

echo "Hello World from my local system!" > sample.txt

3. Stage and commit the file

git add sample.txt

git commit -m "Initial commit with sample file"

4. Link it to your GitHub repository (Create an empty repo on GitHub first to get this URL)

git remote add origin https://github.com/your-username/your-repo-name.git

5. Rename your default branch to main and push

git branch -M main

git push -u origin main

4. **What is the difference between a Fork and a Pull Request? Answer in 2 lines.**

ANS:

A Fork is an independent, personal copy of someone else's GitHub repository created under your own account, allowing you to experiment freely without affecting the original project. A Pull Request (PR) is a formal proposal sent from your forked repository back to the original project owner, asking them to review and merge your new changes into their main codebase.

5. **Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.**

ANS:

1. Ensure you are on the main branch and it's up to date

git checkout main

git pull

2. Create and switch to the new branch named feature-test

git checkout -b feature-test

3. (Make your changes to a file here) then stage the modified file

git add .

4. Make your commit on the feature-test branch

git commit -m "Add test feature modifications"

5. Switch back to the main branch

git checkout main

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6. Merge the feature-test branch into main

git merge feature-test

7. (Optional) Delete the local feature branch once successfully merged

git branch -d feature-test